

_____ employs a different modeling and solution logic than linear programming

- ▶ Transportation Model
- ▶ Inventory Control Model
- ▶ **Dynamic Programming**
- ▶ None of the above

To identify and maintain the proper precedence relationship between activities those are not connected by event, we introduce

- ▶ Parallel Activity
- ▶ **Dummy Activity**
- ▶ Sequential Activity
- ▶ None of the above

EST and EFT of activities are calculated in

- ▶ **Forward pass**
- ▶ Backward pass

Critical path is obtained by connecting the jobs having

- ▶ Activities having same EST and LST
- ▶ Activities having same EFT and LFT
- ▶ Activities having zero slack
- ▶ **All of the above**

In LP problems Additivity means that

- ▶ **The effect of two different programs of production is the same as that of**

a joint program

► The doubling (or tripling) the product will exactly double (or triple) the profit and the required resource

- Both (a) & (b)
- None of the above

Two of the first steps of OR process encompass the actual use of OR techniques.

These steps are

- **Model Construction and Model Solution**
- Observation and Implementation
- Definition of the problem and Model Solution
- Model Solution and Implementation of results

Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, then which relation is true

- $TS \leq FS$
- $INDS \leq FS$
- $FS \leq TS$
- **Both (b) & (c)**

Best possible time estimate that a given activity would take under normal conditions which often exist, is called

- **Most Likely time estimate**
- Pessimistic time estimate
- Smallest time estimate

- ▶ None of the above

Standard Deviation S.D is

- ▶ **One sixth of the difference between pessimistic time estimates and optimistic time estimates**
- ▶ One sixth of the difference between pessimistic time estimates and most likely time estimates
- ▶ One sixth of the difference between optimistic time estimates and most likely time estimates
- ▶ One sixth of the difference between most likely time estimates and optimistic time estimates

EST and EFT of activities are calculated in

- ▶ **Forward pass**
- ▶ Backward pass
- ▶ Path does not effected

_____ may be less than most likely time estimate

- ▶ **Pessimistic time estimate**
- ▶ Smallest time estimate
- ▶ Optimistic Time estimate

The dummy activities consume

- ▶ **No time, no resources**
- ▶ No time but some resources
- ▶ Some resources in minimum time

- ▶ None of these

If an activity consumes no time and no resources then this activity is called

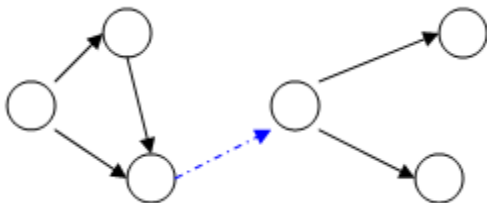
_____.

- ▶ **dummy activity**
- ▶ sequential activity
- ▶ critical activity
- ▶ cyclic activity

Cost period = ----- × (No of ordered items)

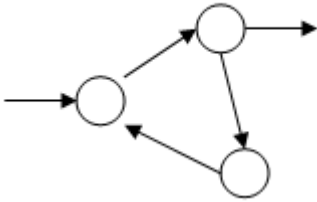
- ▶ Holding cost
- ▶ Set up cost
- ▶ Stock out cost
- ▶ **Item cost**

The following network is an example of



- ▶ Redundancy
- ▶ Cycling
- ▶ Looping
- ▶ **Merging**

The following network is an example of



- ▶ **Redundancy**
- ▶ Dangling
- ▶ Cycling
- ▶ Dummy

Which one is best describe Micro Economic Planning?

- ▶ Distribution of fertilizer
- ▶ **Improving the layout of a workshop in a company**
- ▶ Investment planning of the country
- ▶ PERT

If $t_0 = 6$, $t_m = 12$ and $t_p = 18$, then $V_t =$ _____

- ▶ 12
- ▶ 2
- ▶ **4**
- ▶ 144

Which inventory model also known as a saw tooth model?

- ▶ **Purchasing Model with no shortages**
- ▶ Purchasing Model with shortages
- ▶ Manufacturing Model with no shortages

- ▶ Manufacturing Model with shortages

For an activity if optimistic time, most likely time estimate and pessimistic time estimate are 3, 6 and 15 respectively then expected time is

- ▶ 4
- ▶ 3
- ▶ **7**
- ▶ 20

A dummy activity is a simulated activity of sorts, one that is of _____ duration and is created for the sole purpose of demonstrating a specific relationship and path of action on the arrow diagramming method.

- ▶ **Zero**
- ▶ Minimum
- ▶ Maximum
- ▶ Average

Activity definition refers to the process of parsing a project into a number of individual tasks which must be completed _____ the deliverables can be considered completed. Activity definitions rely on a number of specific input processes.

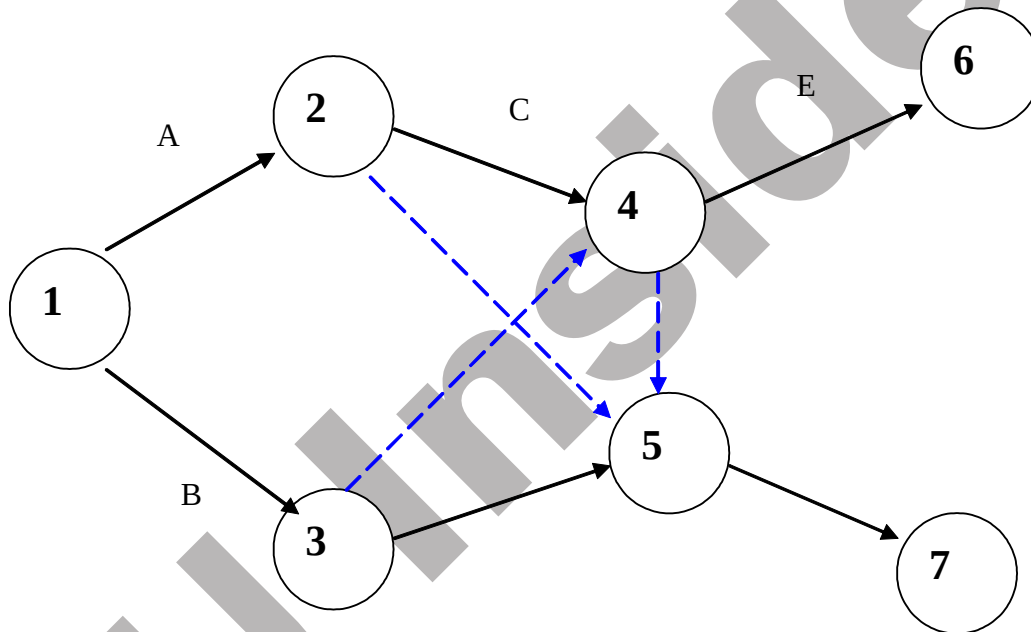
- ▶ **before**
- ▶ both before and after
- ▶ after

A forward pass is used to determine and calculate the _____ dates,

through utilization of a previously specified start date.

- ▶ **early start and early finish**
- ▶ late start and early finish
- ▶ early start and late finish
- ▶ late start and late finish

Is this Network legal?



▶ Yes.

▶ **No.**

$K = Z \times (-----)$

Where K is called service factor.

▶ $\sqrt{\pi/2}$

▶ $\sqrt{2/\pi}$

▶ $\sqrt{2\pi/3}$

▶ $\sqrt{3\pi/2}$

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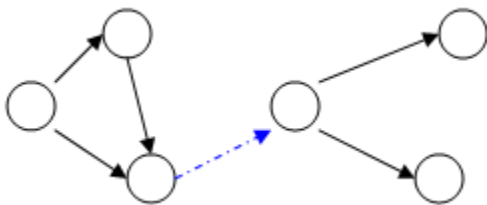
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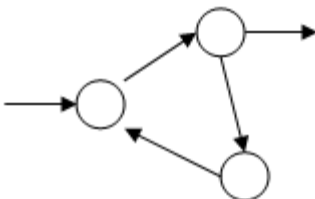
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For backward pass computations

- ▶ Earliest start time $\geq$ Latest start time
- ▶ Earliest start time $\leq$ Latest start time
- ▶ Earliest start time $+$ Latest start time = 0
- ▶ Earliest start time $=$ Latest start time

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In a quadratic programming problem unlike linear programming problem

- ▶ Only objective function is quadratic
- ▶ Both objective function and constraints are quadratic
- ▶ Only constraints are quadratic
- ▶ At least one of objective function or constraint must be quadratic

Solution region of the constraint $x \geq 0$ is

- ▶ Half plane to the right of straight line $x = 0$
- ▶ Half plane to the right of y-axis
- ▶ Half plane to the region where abscissas are non-negative
- ▶ All are equivalent

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Total cost per period = Item cost + Order cost + Holding cost + _____.

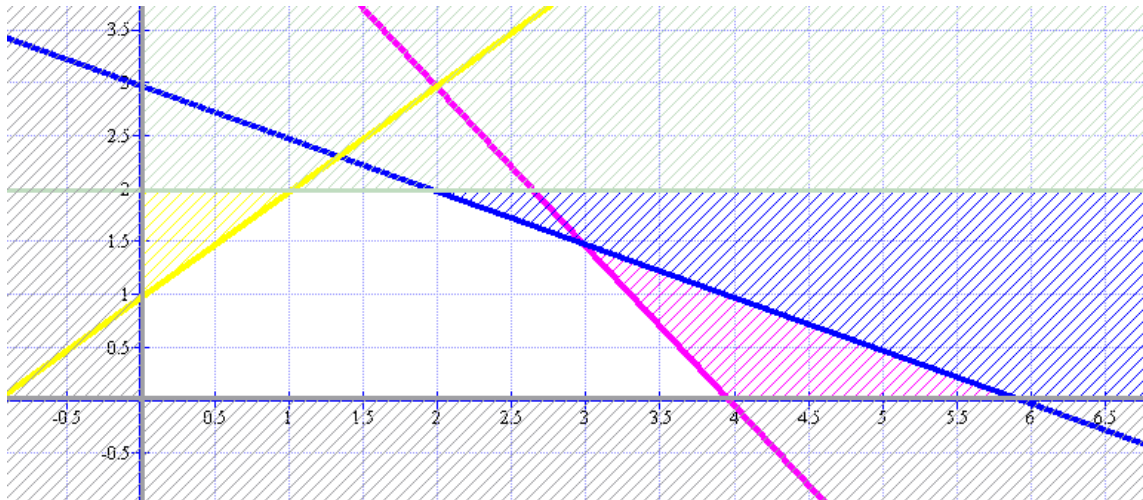
- ▶ Shortage cost
- ▶ Optimum Shortage (S^*)
- ▶ Economic Order Quantity. (Q^*)
- ▶ Maximum Inventory. ($I_{\max}$)

Questions:

MTH601

Mid Term

Question No: 1 (Marks: 2)



Using above graph, Write down the equation (Blue shaded area).

Question No: 2 (Marks: 3)

Draw a net work diagram for the following dependency relationship.

A controls F, G and H; B controls G and H with H controlled by C.

Question No: 3 (Marks: 5)

Draw the graph of the following constraints.

$$\begin{aligned} x &\leq 4 \\ 2y &\leq 12 \\ 2x + 3y &\leq 18 \\ x \geq 0, \quad y &\geq 0 \end{aligned}$$

Question No: 4 (Marks: 10)

Solve by Graphical method:

Shows corner points and optimize the objective function.

$$\text{Max. } Z = 5x_1 - 3x_2 + 10$$

Subject to :

$$2x_1 - 3x_2 \geq 6$$

$$x_1 + 4x_2 \leq 10$$

$$2x_1 + 5x_2 \geq 4$$

$$x_1 \geq 0, x_2 \geq 0$$